WANCONG ZHANG

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Education

PhD in Computer Science, New York University, Courant Institute *09/2023-Present*
Advisor: Prof. Yann LeCun

Research / Employment

Research Scientist Intern, Advanced Machine Intelligence *05/2026-Present*
- Hierarchical JEPA world models for long horizon planning. Paper

PhD Candidate, NYU
Advised by Prof. Yann LeCun *09/2023-Present*
- Joint embedding prediction architecture (JEPA) for zero-shot planning. Paper
- Designed a probing benchmark for unsupervised visual representations in RL, improving Atari pretraining performance beyond SOTA. Paper

Research Scientist Intern, Meta (FAIR)
Advised by Dr. Nicolas Ballas *05/2025-01/2026*
- Hierarchical world models for long horizon planning. Paper

Senior Deep Learning Researcher, AssemblyAI *02/2021-08/2023*
- Spearheaded the development of industry-leading automatic speech recognition (ASR) models (Conformer-1, Conformer-2, realtime ASR).
- Directed teams of up to 10 researchers/engineers, deploying multi-node training pipelines at scale (Google Cloud + on-prem clusters).

M.S. Researcher, NYU
Advised by Profs Sam Bowman & Rajesh Ranganath *09/2019-06/2020*
- Proposed MixUp algorithms for sentence classification using transformers, reducing calibration error by 50%. preprint
- Built customized, fused CUDA kernels for a novel gateless RNN architecture. Blog

Research Intern, University of Toronto
Advised by Prof. Marzyeh Ghassemi *06/2019-09/2019*
- Multimodal representation learning (labs + clinical notes) for healthcare. Paper

Peer Reviewed Publications

Hierarchical Planning with Latent World Models
W Zhang, B Terver, A Zholus, S Chitnis, H Sutaria, M Assran, R Balestriero, Y LeCun, N Ballas
The Conference on Robot Learning (**CoRL**), 2026 [Paper]

Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models
W Zhang*, V Sobal*, K Cho, R Balestriero, T Rudner, Y LeCun
Advances in Neural Information Processing Systems (**NeurIPS**), 2025 [Paper]
Best Paper Award, ICML 2025 Workshop on Building Physically Plausible World Models

Light-weight probing of unsupervised representations for Reinforcement Learning
W Zhang, A Chen, V Sobal, Y LeCun, N Carson
Reinforcement Learning Conference (**RLC**) 2024 [Paper]
A Comprehensive EHR time series pre-training benchmark
M McDermott, B Nestor, **W Zhang,** P Szolovitz, A Goldenberg, M Ghassemi
The ACM Conference on Health, Inference, and Learning (**CHIL**) 2021 [Paper]

Preprints

H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning
W Zhang*, B Terver*, M Rabbat, Y LeCun, R Balestriero
arXiv preprint 2026 [arXiv]

Conformer-1: Robust ASR via Large-Scale Semisupervised Bootstrapping
W Zhang, L Chkhetiani, F Ramirez, Y Share, A Vanzo, M Liang, S Martin, G Oexle, R Bousbib, T Pesach, M Nguyen, D Pulliam, D Donato
arXiv preprint 2024 [arXiv]

Mixup training leads to reduced overfitting and improved calibration for the transformer architecture
W Zhang, I Vaidya
arXiv preprint 2021 [arXiv]

Teaching

Graduate Student Instructor: Deep Learning (NYU) *Fall 2024*

- Designed and hosted the final course competition for ~150 students (project page)
- Designed homework assignments and coordinated graders
- Supervised M.S. students with independent research projects on video SSL